

Manifold-Informed Architecture Benchmark

MNIST: 10 classes, 784D input

Generated: 2026-04-14 20:54:27 | Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD

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Best: Standard (128→64) test acc = 0.9742 ± 0.0010

Manifold: 784D → 27D (96.6% noise dimensions)

Experimental Setup

Parameter	Value
Dataset	MNIST
Input dimensionality	784D
Classes	10
Intrinsic dim (d)	27
Variance threshold (τ)	0.9
Epochs	50
Trials	5
Batch size	128
Learning rate	0.001
k neighbors (local PCA)	50

Manifold Discovery

Local PCA over the training set, $k = 50$ neighbors. Intrinsic dimensionality d is taken as the maximum per-class maximum at $\tau = 0.9$, clamped to $n_{\text{classes}} = 10$.

τ	Mean d	Std	Min	Max	Noise (%)
0.95	29.7	2.9	17	34	96.2
0.90	22.2	2.8	10	27	97.2
0.85	17.4	2.5	7	22	97.8
0.80	14.1	2.3	5	18	98.2

Per-Class Intrinsic Dimensionality

Class	Mean d	Std	Min	Max
Digit 8	24.8	1.0	23	27
Digit 3	24.2	1.5	18	26
Digit 5	23.8	2.0	16	27
Digit 4	23.3	1.1	20	25
Digit 2	23.0	3.4	7	27
Digit 0	22.9	1.4	18	25
Digit 9	20.6	2.0	12	23
Digit 6	20.6	2.9	6	24
Digit 7	19.2	2.8	11	23
Digit 1	16.9	1.5	13	20

Architecture Comparison

Architecture	Params	Test Acc ($\mu \pm \sigma$)	Loss	Acc/Kparam
Euclidean KNN (k=7)	0	0.8948 ± 0.0000	—	—
ManifoldModel ($\tau=0.9$)	0	0.8958 ± 0.0000	—	—
Standard (128→64) ★	109,386	0.9742 ± 0.0010	0.3279	0.0089
Wide Manifold (4d→2d→d, d=27)	92,431	0.9735 ± 0.0012	0.2989	0.0105
Manifold (2d→d, d=27)	44,155	0.9677 ± 0.0006	0.3700	0.0219
PCA→27D + MLP (2d→d)	3,277	0.9623 ± 0.0028	0.1430	0.2937
Intrinsic Dim (PCA→27D→output)	1,036	0.9511 ± 0.0012	0.1655	0.9181

★ = best performing architecture

Key Findings

- **Best architecture:** Standard (128→64) — test accuracy 0.9742 ± 0.0010
- **Manifold compression:** 784D → 27D (96.6% of ambient dimensions carry no structure)

Result Figure

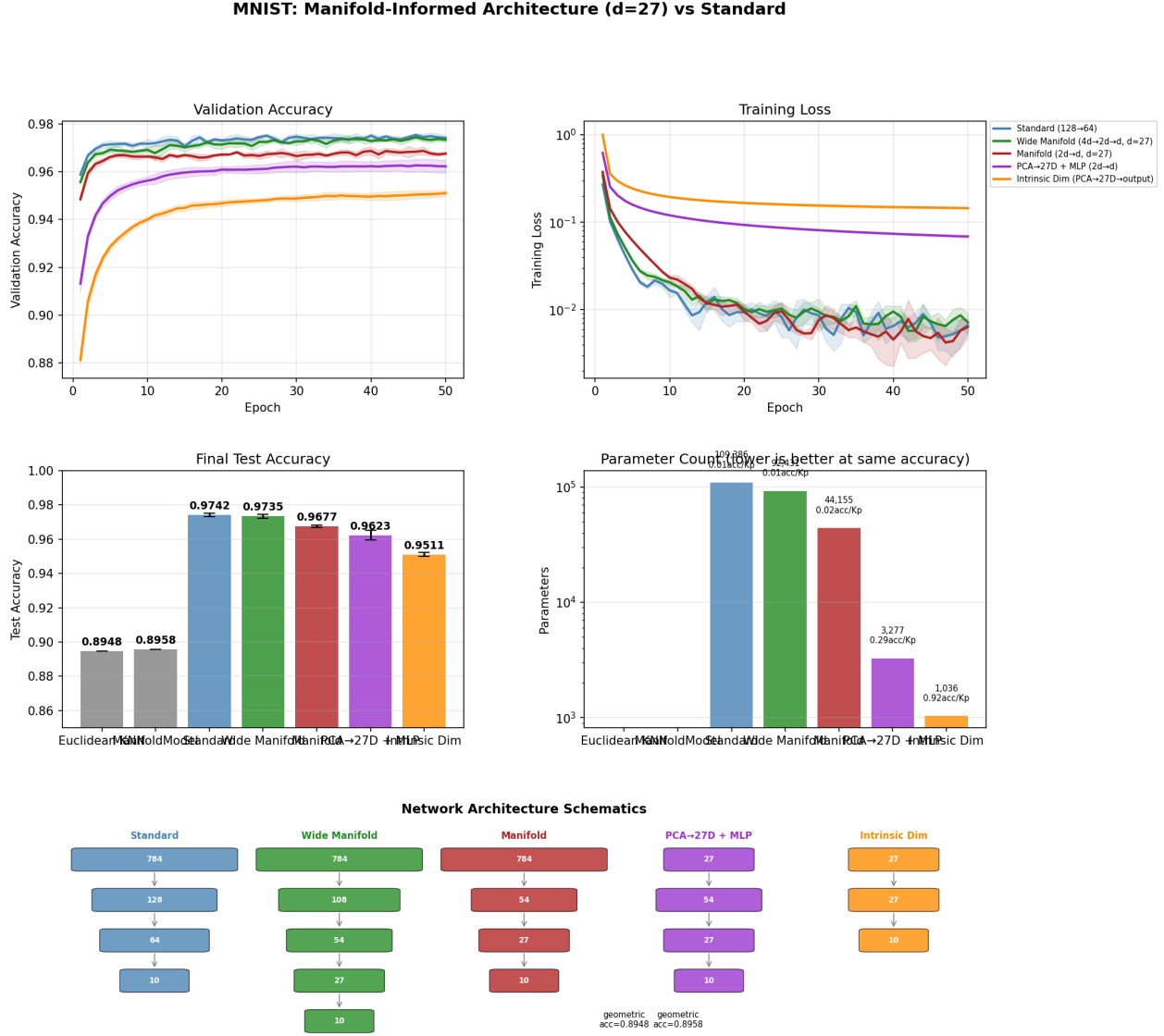


Figure 1: MNIST manifold-informed vs. standard architecture comparison. $d = 27$, 50 epochs, 5 trials.

Provenance

Item	Value
Repository	Flux-Frontiers/waverider
Commit	4b8002e (-abbrev-re 4b8002ee9a2e3d56a219d7dab695a80b8efd1e07)
Commit date	2026-04-14 20:51:52 -0400
Commit message	add: cifar10 results
Python	3.12.13
TensorFlow	2.16.2
Device	CPU (forced)
Host	Turing
OS	macOS-26.4-arm64-arm-64bit
Machine	Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD
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